Kieran van der Heijde

Python Engineer

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SUMMARY

Self-taught Python engineer at ASML who builds and runs production systems alone: a ranked gaming platform with 2,600+ users, a sandbox that runs untrusted student code, and Angelo, an open-source mutation tester in Rust that runs Python mutation testing 10.8× faster without changing the score. Gave a guest lecture on it at Fontys.

SKILLS

Languages: Python, SQL, Rust, JavaScript, C#, HTML/CSS

Backend: FastAPI, Flask, SQLAlchemy, Pydantic, REST APIs, Jinja2, HTMX

Data & testing: PostgreSQL, pandas, NumPy, Metabase, pytest, mutation testing

Infrastructure: Docker, gVisor, Traefik, Dokploy, Linux, Git, CI/CD

EXPERIENCE

ASML, Veldhoven, Netherlands, Feb 2025 - Present

Python Engineer, Nov 2025 - Present

- Build and extend internal Python and JavaScript tools used across multiple business units.
- Supervised my first intern, a Fontys student, through a six-month internship. We held weekly check-ins while I worked one day a week. He built and optimised an internal information website and finished with a 10/10 from Fontys.

Automation Developer, Feb 2025 - Nov 2025

- Replaced manual spreadsheet processes with Python automation, which led to development projects for the team.

PROJECTS

Angelo, creator, github.com/kvandeh/Angelo

- Cut a 200-mutant run from 48.1 s to 4.5 s with an identical score. Batch mutation packs mutants that no single test reaches together, using per-test coverage (4.6× faster on its own). Test selection and warm pytest workers add the rest.
- Made speed impossible to buy with accuracy: added guards so that 0 matched mutants is an error, not a pass, and every batch survivor is re-run on its own. Validated on flask and click.
- Finds lines that coverage counts as tested but that no test checks. Ships as one binary with a CI gate and SonarQube reporting.

PeekSeek, creator and sole engineer, peekseek.net

- Ranked matchmaking platform for Unturned, officially partnered with the game's studio, Smartly Dressed Games. 2,600+ users and 10,000+ matches.
- Run the whole system alone: backend, PostgreSQL, a Discord bot and game servers, deployed as Docker containers with Dokploy on one dedicated server.
- Built the analytics in Metabase with native SQL on PostgreSQL: growth panels, match-fixing detection, player co-occurrence and account-linkage analysis.

AlgoStudio, sole engineer, algostudio.io

- Platform for teaching programming. Students write Python or C# and get live feedback per test. Teachers generate problems and test cases with an AI assistant. Ships with a VS Code extension.
- Runs untrusted code in throwaway Docker containers under gVisor: no network, capped memory, CPU and processes, and a warm pool for speed. FastAPI and Flask on PostgreSQL, with Jinja2 and HTMX.

SPEAKING & TEACHING

Guest Lecturer, Mutation Testing
Fontys University of Applied Sciences, Eindhoven, Netherlands, Oct 2026
Taught mutation testing to third-year ICT students. It was the first ICT guest lecture there that year, and I was invited back.

Swim Instructor & Coach (volunteer)
Helmondse Watervrienden, Helmond, Netherlands, Feb 2021 - Present
Teach and coach swimmers of all ages.

EDUCATION

BSc Communication & Information Sciences
Tilburg University, Tilburg, Netherlands, Sep 2026 - Expected 2029

HAVO, Natuur & Techniek and Natuur & Gezondheid (bilingual)
Jan van Brabant College, Helmond, Netherlands, 2024

LANGUAGES

Dutch (native) and English (fluent, bilingual secondary education, Cambridge IGCSE English 2024).